Driving Score Predictor

CSE-MBMU

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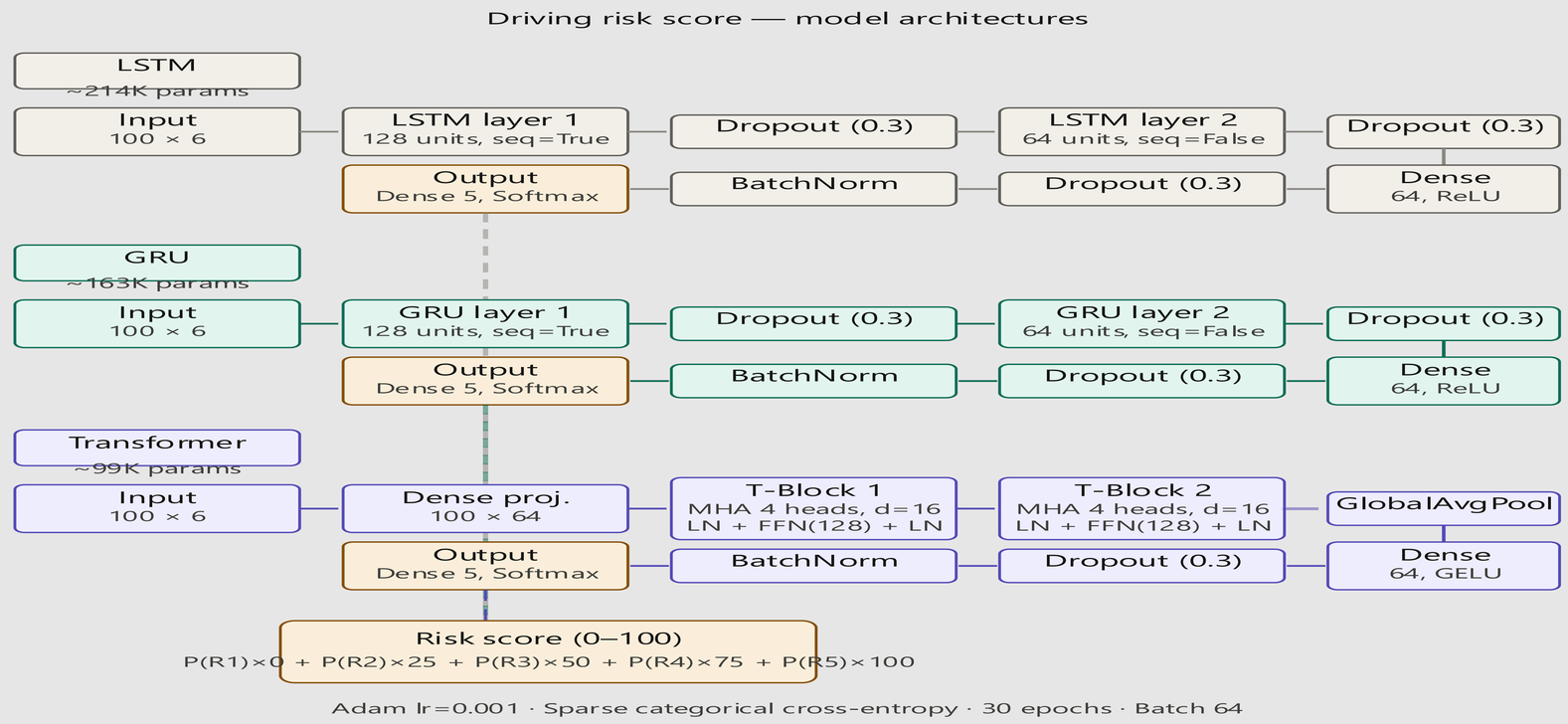
Introduction

- Road safety on two-wheelers is a critical concern in India. This project automatically assesses driving behaviour using only a smartphone's built-in accelerometer and gyroscope — no GPS or cameras required.
- Data collected using the Android sensor app. 2.98M readings from 23 riders across 193 sessions covering multiple vehicles, speeds, and phone placements.
- Three deep learning models — LSTM, GRU, and Transformer — trained on 100-row sliding windows. Output is a 0–100 Driving Risk Score with star rating (★ to ★★★★★).
- Best result: GRU achieves 92.16% validation accuracy and AUC-ROC of 0.9925 across 54,850 windows.

Dataset Description

- Collected using Android IMU sensor app at ~100 Hz. Each session rated 1 (most dangerous) to 5 (safest) by folder labelling. Columns: Timestamp, X_Acc, Y_Acc, Z_Acc, X_Gyro, Y_Gyro, Z_Gyro.
- Raw: 2,987,342 rows | After cleaning: 2,752,364 (92.13% retention) | Windows: 54,850 | Split: 80/20 stratified
- Variations: 23 riders, scooter + bike, slow/medium/fast speeds, pocket/handlebar/backpack placements. Rating distribution: 1★ 39.6%, 2★ 14.1%, 3★ 17.9%, 4★ 18.9%, 5★ 9.5%

Network Architecture Diagram

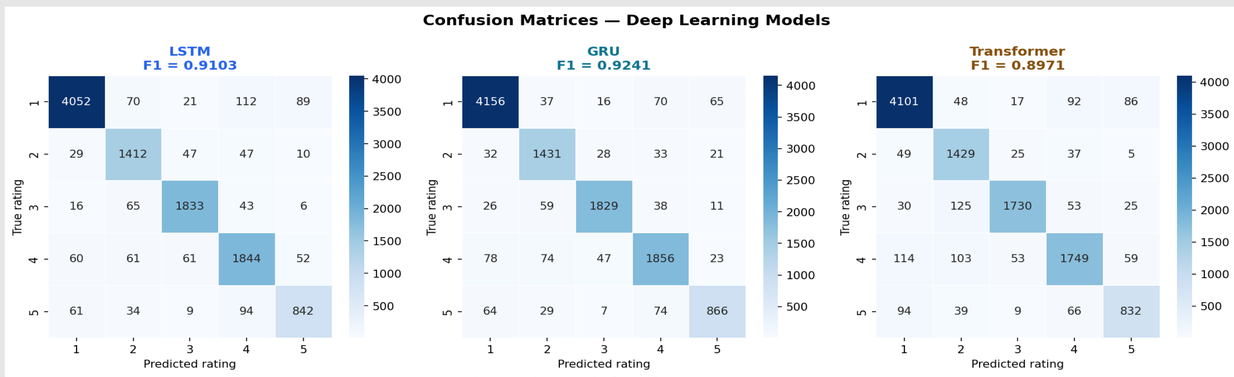


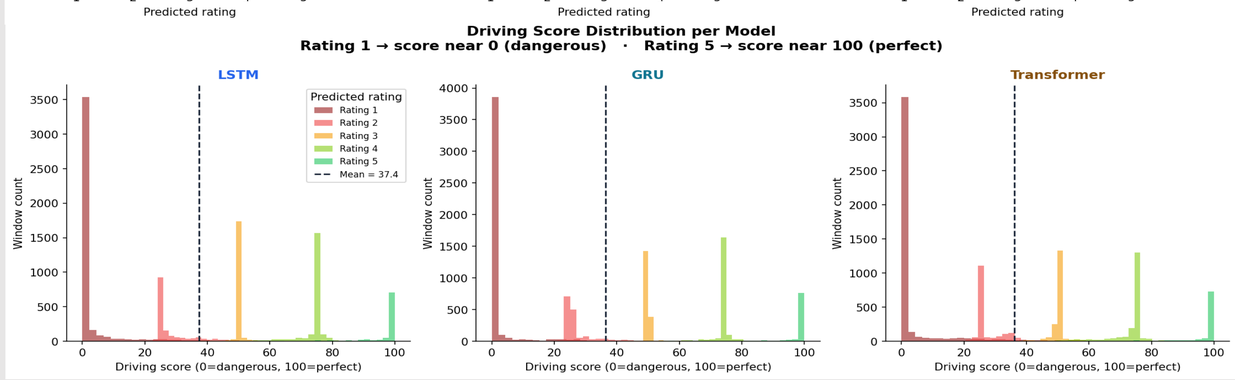
QUICK INSIGHTS:

1. Total Parameters:
   LSTM: ~214,000
   GRU: ~163,000
   Transformer: ~99,000
2. Optimizer: Adam (lr = 0.001)
   Loss Function: Sparse Categorical Cross-Entropy
   Library: TensorFlow 2.16 / Keras
3. Inference Requirements:
   Memory: ~50 MB (GRU model)
   Inference time on CPU: ~2 ms per window [Intel Core i5 or equivalent]

Model Training

- Input preprocessing: 100-row sliding window (50% overlap), z-score normalisation per axis using training statistics only. Labels 1–5 shifted to 0-indexed.
- Train/Test: 80/20 stratified split. Train: 43,880 windows. Test: 10,970 windows. Batch size: 64. Epochs: 30 (no early stopping — all models trained identically for fair comparison).
- Optimizer: Adam (lr=0.001). Loss: sparse categorical cross-entropy.





Model Validation and Discussion

- All three models evaluated on the held-out test set (10,970 windows) using weighted F1, AUC-ROC (OvR), precision, and recall.
- Results: GRU — F1: 0.9241, AUC: 0.9925 | LSTM — F1: 0.9103, AUC: 0.9903 | Transformer — F1: 0.8971, AUC: 0.9882
- GRU outperforms Transformer because driving behaviour unfolds gradually over time — sequential step-by-step processing captures local temporal dependencies better than global attention on 100-row windows.

Limitations and Future Scope

- Dataset is imbalanced — Rating 1 has 39.6% of windows vs Rating 5 at only 9.5%. Future work: apply class weighting or SMOTE oversampling.
- Model trained on specific riders and roads — generalization to unseen riders in different cities not fully validated. Future: leave-one-rider-out cross-validation on larger cohort.
- Currently offline inference only. Future: deploy lightweight GRU as on-device TFLite model for real-time scoring during the ride.

Project Repositories

Scan for Dataset

Scan for Source Code

Note: Trained model weights are also available in the source code repository.

Reviewed and Edited By: Abhisek Gour, Assistant Professor Department of Computer Science and Engineering, MBM University, Jodhpur